

Total No. of Questions : 6]

SEAT No. :

P209

[Total No. of Pages : 3

APR - 17/TE/Insem. - 45

T.E. (Information Technology)

SYSTEMS PROGRAMMING (314450)

(2012 Course) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) For the following piece of assembly language code, show the contents of symbol table, literal table and pool-tab. Assume size of instruction equal to one. **[6]**

```
START 510
LOOP MOVER AREG, A
      MOVER AREG, =5'
      MOVEM AREG, A
      BC GT, NEXT
      MOVER CREG, B
      ADD CREG, = '1'
      MOVEM CREG, B
      SUB CREG, =5'
      LTORG
      ADD CREG, B
      BC LE LOOP
NEXT SUB AREG, = '1'
      BC LT, BACK
      LTORG
      STOP
      ORIGIN 525
      MULT CREG, =1'
      A DC 1
BACK EQU LOOP
      B DS 15
      END
```

P.T.O.

- b) Explain how a single pass assembler works for the following example[4]

```
START 300
READ A
LOOP MOVER AREG ,A
      SUB AREG, ='1'
      BC GT, LOOP
STOP
A DS 1
END
```

OR

- Q2) a) For the following assembly language code, show the contents of macro name table, macro definition table. Finally write down the code after macro expansion. Is the use of stack necessary for macro expansion? Justify. [6]

```
MACRO
INCR &M,&I=,&REG=AREG
MOVER &REG,&M
ADD &REG,&I
MOVEM &REG,&M
MEND
MACRO
CALC &X,&Y=B,&OP=MULT
MOVER AREG,&X
&OP AREG,&Y
MOVEM AREG,&X
MEND
START 100
READ N1
CALC A,OP=ADD
ADD AREG,21
LDA CREG,100
SUB CREG A
INCR B,REG=B,I=A
A DS 1
B DS 1
END
```

- b) Define Assembler. List and explain the databases used in Pass I and Pass II of two pass assembler. [4]

- Q3) a) Define Loader. List basic functions of loader and explain how they are performed in absolute loader scheme. [6]

- b) What is meant by Overlay structure? Discuss with examples. [4]

OR

- Q4)** a) Give the various data structures in the design of pass-1 and pass-2 of a Two-pass direct linking loader for the given example. [6]

	Rel. Addr.
PGA START	0
ENTRY PGA1	10
EXTRN PGB	20
DC A(PGA), A(PGB+4)	30
PGA1 DC A(PGA1-PGA)	34
END	
PGB START	0
ENTRY PGB1	5
EXTRN PGA	10
PGB1 DC A(PGB1)	14
PGB2 DC A(PGB+4), A(PGB1-PGB)	18
PGB3 DC A(PGB-PGA-16)	22
END	

- b) Explain 'Compile and Go' loader scheme. What are the advantages and disadvantages of this scheme? [4]

- Q5)** a) What is a compiler? What are the various phases in the design of a compiler? [6]

- b) Convert the given RE to DFA. [4]
 $(a+b)^* . (a+\epsilon)^*$

OR

- Q6)** a) Perform Lexical analysis for the given code. [6]

```
Void main()
{
    float ch1, ch2;
    int sno;
    clrscr();
    scanf("%f %d",&ch1,&sno);
    ch2 = ch1 + sno;
    sno++;
    printf("%f", ch2);
    getch();
}
```

- b) Explain LEX Specification and Generation of Lexical Analyzer by LEX. [4]

✕

✕

✕